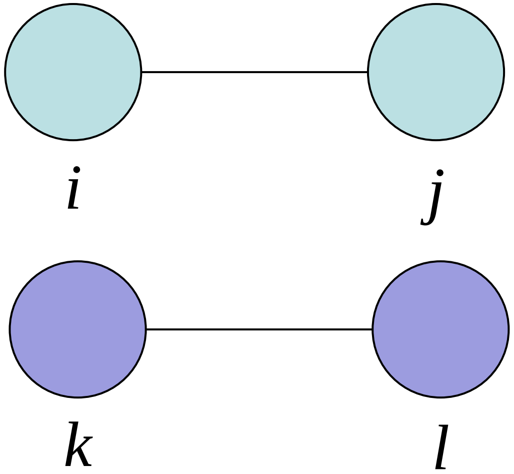
The research frontiers of network dynamics & Introduction to RSiena

Networks and Innovation Training Programme
Dr Antoine Vernet & Dr Anne ter Wal

September 13, 2013



Q

In many networks we observe
that similar others are connected.
How come?

Social selection versus social influence

Social selection | **Social influence**

Only a longitudinal analysis can disentangle these two forces in the formation of networks

People tend to connect to similar others.

People tend to become more similar to those to whom they are connected.

The NEW ENGLAND JOURNAL of MEDICINE

SPECIAL ARTICLE

The Spread of Obesity in a Large Social Network over 32 Years

Nicholas A. Christakis, M.D., Ph.D., M.P.H., and James H. Fowler, Ph.D.

ABSTRACT

BACKGROUND

The prevalence of obesity has increased substantially over the past 30 years. We performed a quantitative analysis of the nature and extent of the person-to-person spread of obesity as a possible factor contributing to the obesity epidemic.

METHODS

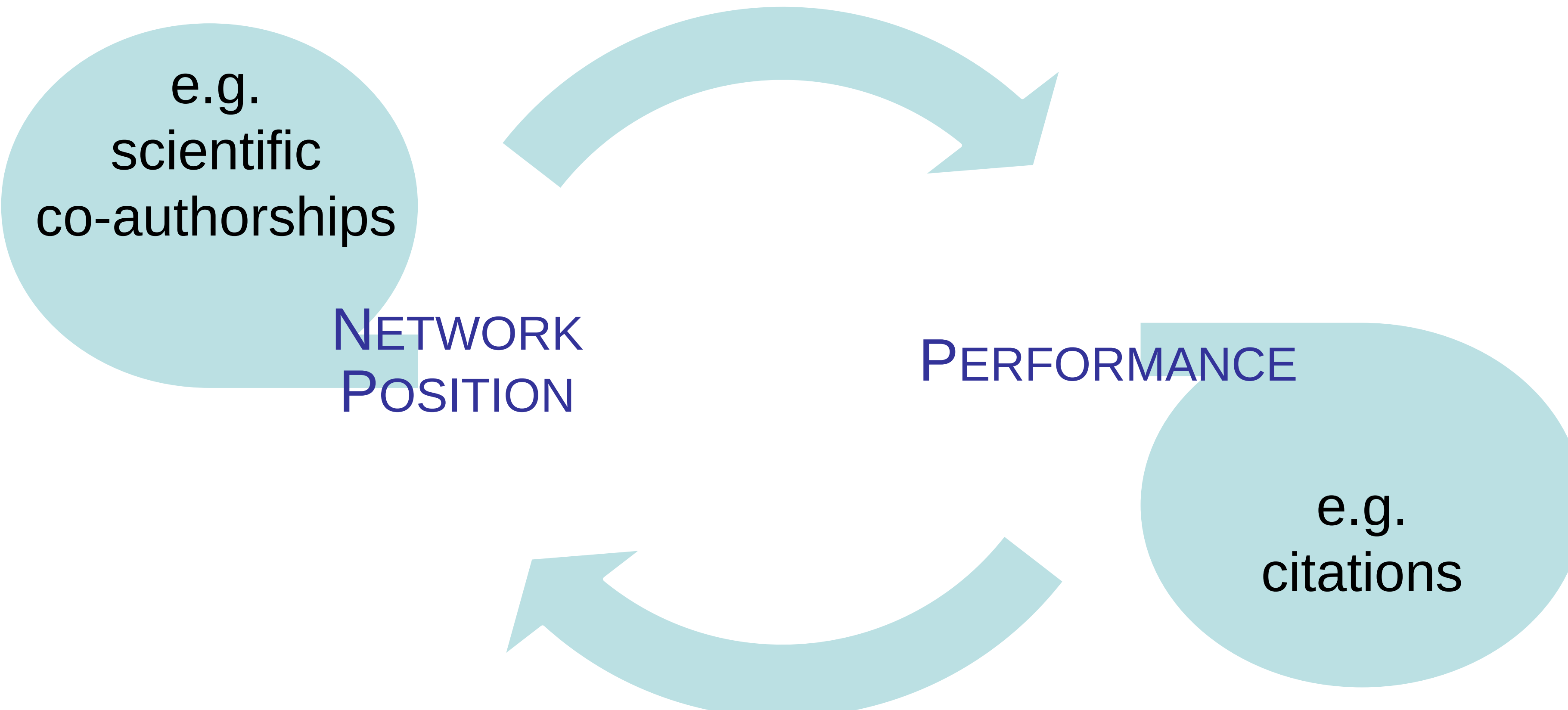
We evaluated a densely interconnected social network of 12,067 people assessed repeatedly from 1971 to 2003 as part of the Framingham Heart Study. The body-mass index was available for all subjects. We used longitudinal statistical models to examine whether weight gain in one person was associated with weight gain in his or her friends, siblings, spouse, and neighbors.

RESULTS

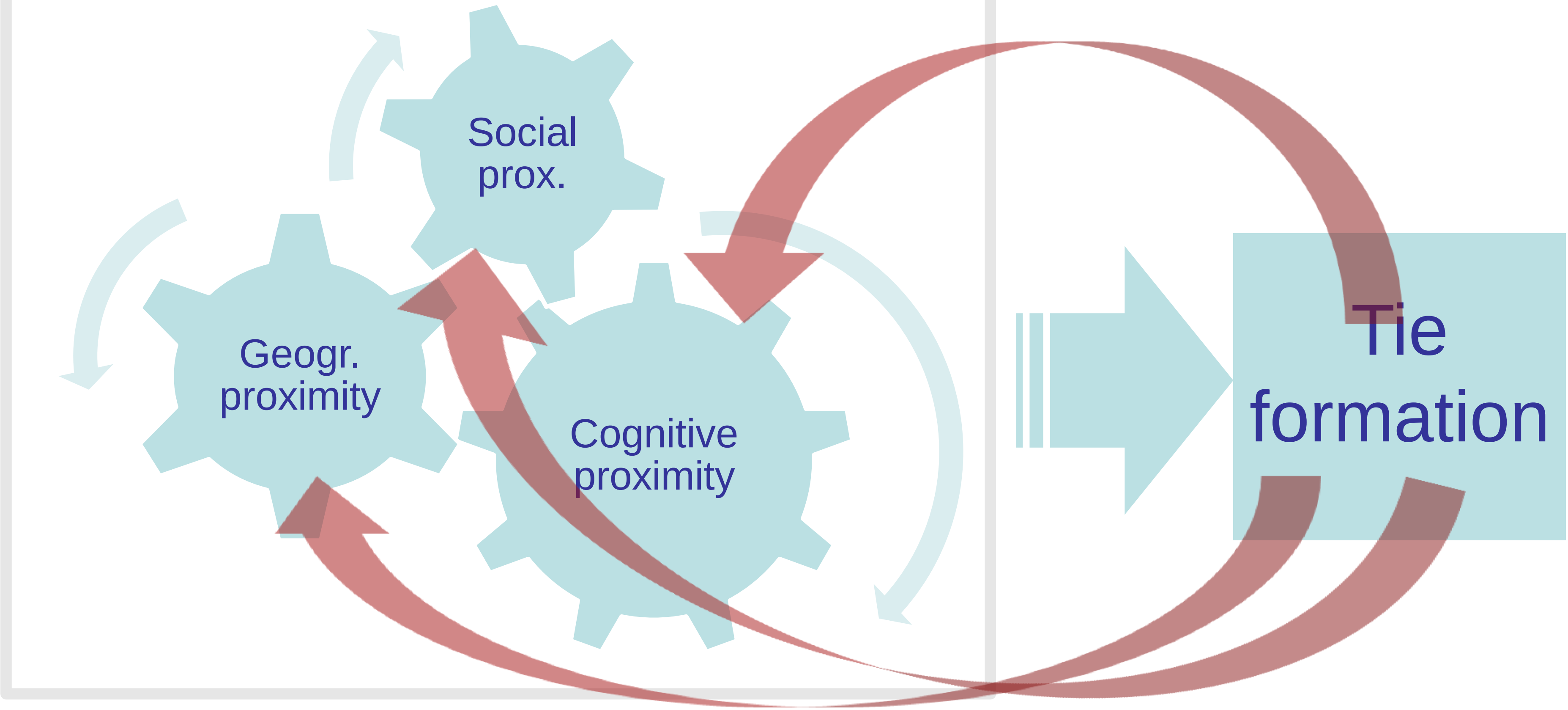
From the Department of Health Care Policy, Harvard Medical School, Boston (N.A.C.); the Department of Medicine, Mt. Auburn Hospital, Cambridge, MA (N.A.C.); the Department of Sociology, Harvard University, Cambridge, MA (N.A.C.); and the Department of Political Science, University of California, San Diego, San Diego (J.H.F.). Address reprint requests to Dr. Christakis at the Department of Health Care Policy, Harvard Medical School, 180 Longwood Ave., Boston, MA 02115, or at christakis@hcp.med.harvard.edu.

N Engl J Med 2007;357:370-9.

Position-performance feedback loop



A dynamic proximity framework



Examples (not using SIENA)

Burkhardt and Brass (1990)

Do powerful people become more central

Or do central people become more powerful?

- Does the diffusion of a new technology (a computerized information system) follow established network patterns with those in power reinforcing their positions or does the diffusion results in changing patterns of interaction and influence?

Lee 2010

Does past performance drive centrality?

Or does centrality boost performance?

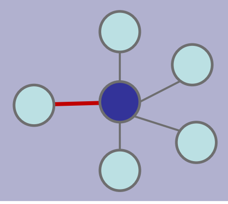
- Inventors with superior track records are more likely to occupy brokerage position. Controlling for past performance and inventor-level fixed effects make the correlation disappear.

Learning objectives

At the end of this lecture and practice session, we hope you will be able to:

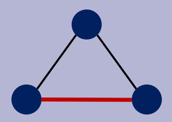
- Categorize **mechanisms of network evolution** and understand how stochastic estimation models can disentangle these.
- Understand the **logic of building a stochastic estimation model**.
- **Conduct** a basic stochastic estimation model in RSiena and **interpret** its outcomes.

Three main drivers of social selection



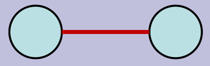
Preferential attachment / Popularity/Activity

The probability that a new node will link to a certain node is proportional to the number of links that node already has.



Transitivity / Triadic closure

The tendency that partners of partners become partners among themselves.



Homophily / (Dis)similarity / Proximity

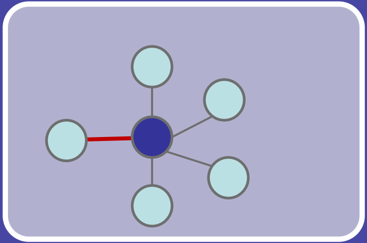
The choice of partner is biased towards partners that are 'similar'.

See also: Powell et al. (2005), Ahuja et al. (2012)

Other structural effects of social selection

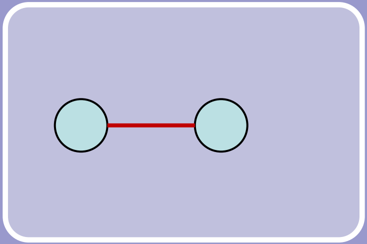
- Density (degree)
  ‘Baseline’ effect that accounts for the observed density in the network and should be included in any model
- Balance
  Tendency to form ties with nodes with a similar degree
- Distances-two
  Tendency for the creation of direct links to alters at geodesic distance 2
- Betweenness effect
  Tendency that a node positions itself between two others that are not yet connected
- Popularity effect
  Tendency in which the number of new links a node gets is proportional to the sum of the (squared) degree of its alters

Drivers of social influence



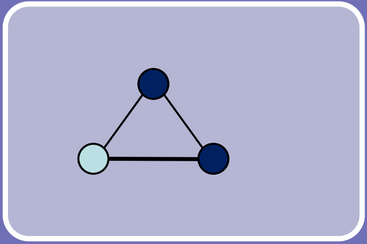
Centrality effects on attributes

The probability that a node changes attribute is proportional to the number of connections it has.



Dyadic social influence

Nodes become more similar to or obtain attributes of those to whom they are connected.



Triadic social influence

Nodes become more similar to or obtain attributes of those with whom they are in a closed triad.

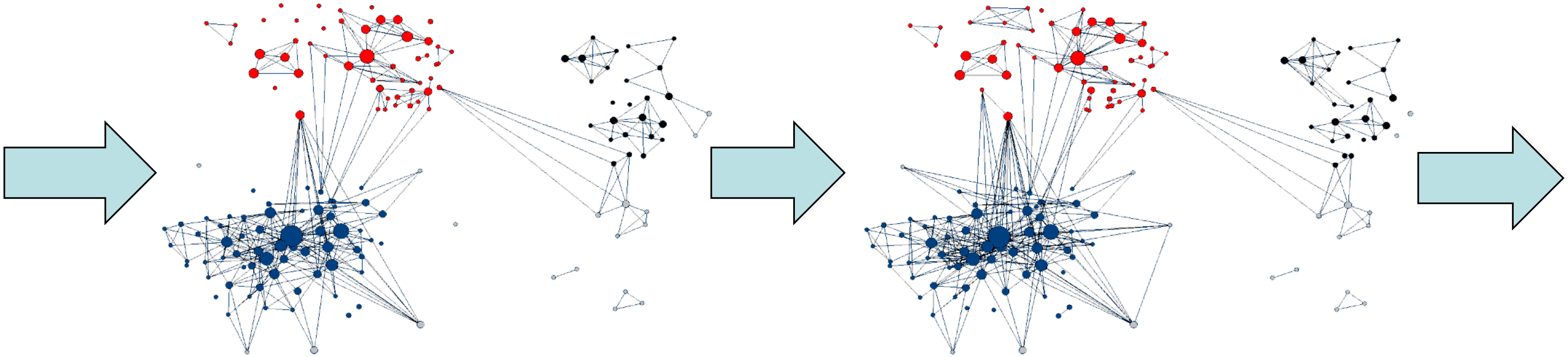
See also: Powell et al. (2005), Ahuja et al. (2012)

UK-INNOVATION RESEARCH CENTRE

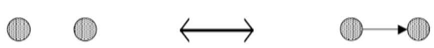
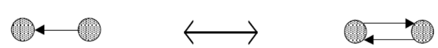
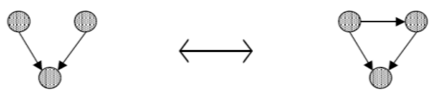
Introduction to SIENA

SIENA - Simulation Investigation for Empirical Network Analysis

- The SIENA software can estimate parameters for drivers of network evolution, distinguishing social selection from social influence.
- It does so by simulating how the network evolved from one state in another (in between two subsequent observations of the network).



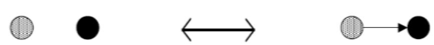
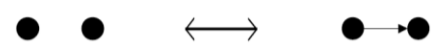
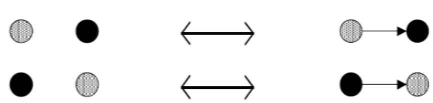
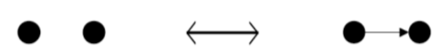
Structural / endogenous effects (social selection)

Out-degree Overall tendency to form ties 	**Reciprocity** Tendency to reciprocate ties 	**In-degree / Popularity** Tendency to form ties to central others 
Transitive triples Tendency towards triadic closure 	**Balance** Tendency to have ties to structurally similar others 	**Three-cycles** Tendency to form relationships in cycles 

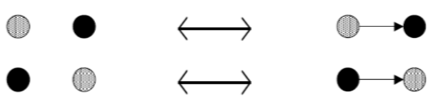
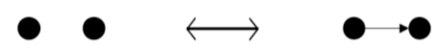
Attribute / exogenous effects

	Actor covariate	Dyadic covariate
Constant covariate	Gender Organization type	Organizational proximity Similarity in educational background
Changing covariate	Performance Opinion	Expertise similarity Level of trust Presence of tie in other network

Attribute / exogenous effects (social selection)

	Actor covariate	Dyadic covariate
Constant covariate	**Covariate ego/alter** Actor attribute determines activity (out-degree) or popularity (in-degree) 	**Covariate similarity** Tendency to form ties to similar others. 
Changing covariate	**Covariate ego/alter** Actor attribute determines activity (out-degree) or popularity (in-degree) 	**Covariate similarity** Tendency to form ties to similar others. 

Attribute / exogenous effects (social influence)

	Actor covariate	Dyadic covariate
Constant covariate		
Changing covariate	**Social influence through degree** The effect of own popularity/activity on attribute 	**Covariate similarity through dyadic social influence** Assimilation of attribute to alters (total or average effect) 

The estimation process

Simulation of network evolution: **three possible actions**

- Creation of a tie
- Dissolution of a tie
- Change of attribute

Estimating the parameters:

- Defining the **target statistics** (observed values)
- Defining initial parameter values
- Iterative search for better parameter values by minimizing the deviation of generated versus observed values.

The estimation is a **stochastic** process. Re-running the simulation does not give the identical outcomes!

The estimation process

Directed and undirected networks

SIENA has originally been developed for analysis of directed networks.

The graphical user interface automatically detects undirected networks and adjusts the set of parameters accordingly.

Undirected networks are more complicated to simulate. Since the creation/dissolution of tie now depends on two actors, the decision rules in the algorithm need to be adapted.

The estimation process

Basic question:

How does the network evolve from one state (t=1) into the next (t=2)?

- Similar to a regression analysis, you select the **parameters** you want to include in the model.
- Since it is unknown in which steps the network has evolved from state into the other, SIENA **simulates** the process of network evolution.
- For undirected networks there are 7 different **algorithms**: decision rules that describe the tie formation process.
- **Markov chain**: the simulation is repeated a 1000 times (default)

Learning objectives

At the end of this lecture and practice session, we hope you will be able to:

- Categorize **mechanisms of network evolution** and understand how stochastic estimation models can disentangle these. ?✓
- Understand the **logic of building a stochastic estimation model**.
- **Conduct** a basic stochastic estimation model in RSiena and **interpret** its outcomes.

Data specification

- Adjacency matrices
  - one for each observation point
  - SIENA uses binary matrices
  - Matrix is dichotomized at a threshold level
  - Default missing value is 9

- Attribute files
  - a single file for constant covariates
  - one file for each observation point (-1) for changing covariates

- Matrix of dyadic covariates
  - Form of an adjacency matrix
  - Integer values between 0 and 255
  - One matrix for each observation point (-1) for changing covariates

Entry and exit of nodes

- **Coding in the adjacency matrix:**
  Use of structural 0s (and structural 1s)

- **A separate file**:
  specifying the times of composition change

Model specification

- **Rate function effects**
  - Basic rate parameter
  - Rate effect for attribute change
  - Covariate's effect for attribute change

- **Tendency effect (for social influence models)**
  - General tendency to change attributes

- **Evaluation / endowment effects**
  - Structural effects
  - Social selection effects
  - Social influence effects

Interpretation of results

Convergence

Good convergence of the simulation is a necessary condition for interpreting the parameter estimates.

t-statistic: the extent to which estimated network properties are in line with observed properties.

Convergence problems:

- Rerunning the simulation with new starting values
- Multicollinearity problems
- Too many weak effects included

Interpretation of results

Parameter estimates

SIENA reports **parameter values** and **standard errors**

t-statistic for significance: parameter estimate / standard error

- Rate parameter
- Density parameter
- Other parameters

Interpretation of results

Multicollinearity

A covariance matrix is provided with correlation coefficients under the diagonal

Statistical tests

- Goodness-of-fit test
- Test the estimated model against a restricted model
- The Newman-Rao test statistic c compares the statistic estimated by the restricted model with the observed statistic
- SIENA reports the test statistic with p-values

Advantages and disadvantages

Advantages:

- Systematic approach of analysing network dynamics
- Possible to detect social influence versus social selection
- Node entry and exit possible

Disadvantages

- The program cannot cope with a too high degree of change
- Size of the networks is limited
- Convergence can be problematic
- Multicollinearity problems can limit the number of effects you want to include
- Valued network analysis not yet implemented
- Two-mode network analysis in progress (Conaldi et al, *ORM*, 2012)

Learning objectives

At the end of this lecture and practice session, we hope you will be able to:

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- Understand the **logic of building a stochastic estimation model**. ?✓
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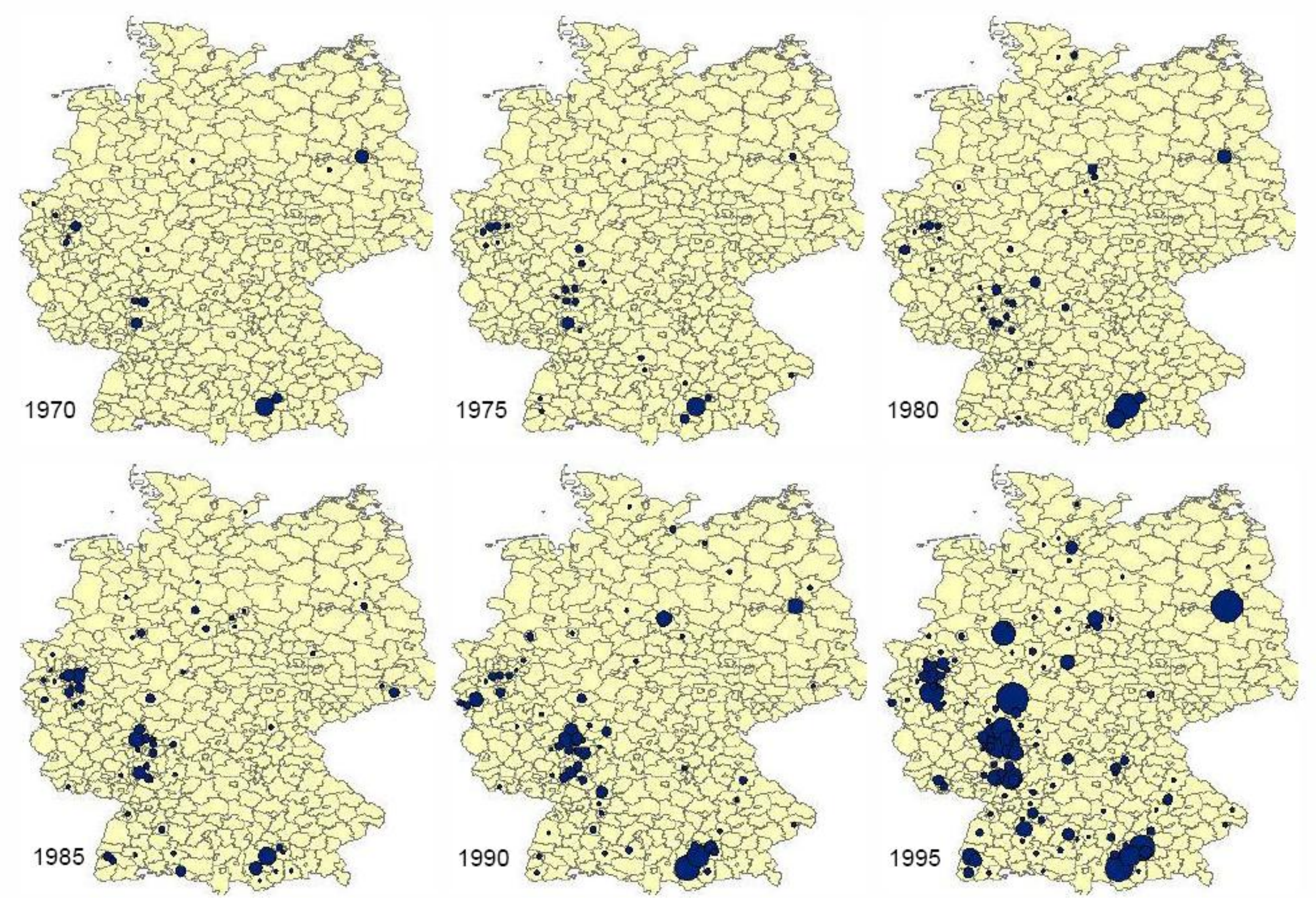
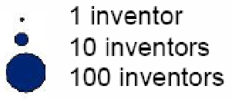
Example German biotech



Number of inventors per district (NUTS3) per year



Source: Ter Wal (2013)

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Example German biotech

What drives the dynamics
of the inventor network
in German biotechnology,
throughout the emergence
and growth of the industry?

Source: Ter Wal (2013)

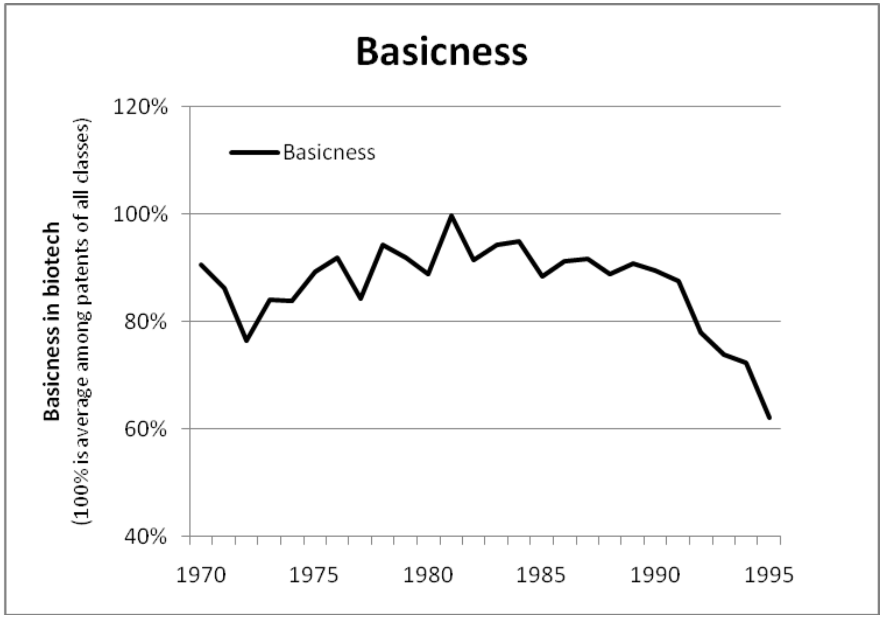
Example German biotech

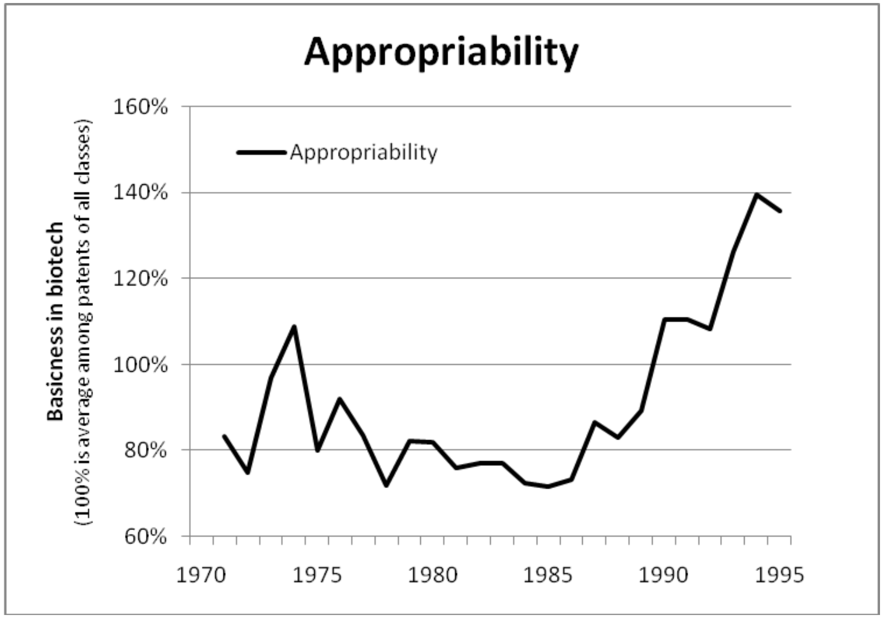
Phase 1 **EMERGENCE**	**Scientific advancements in cellular biology in 1960s and 1970s** **Dominant role of science-based DBFs** **Knowledge is strongly tacit (random screening / trial and error)** **Generic knowledge base / Broad learning strategies**
Phase 2 **GROWTH**	**Large pharmaceutical firms get more involved in the 1980s** **Leading to the development of commercial applications** **Knowledge becomes increasingly codified (rational drug design)** **Specialized knowledge base geared toward specific applications**

Gambardella (1995), Powell et al. (1996), Audretsch (2001), Nesta & Saviotti (2006), Gilsing & Nooteboom (2006)

Source: Ter Wal (2013)

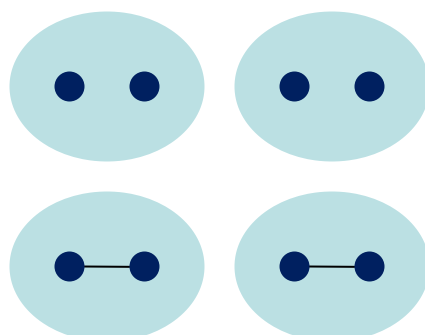
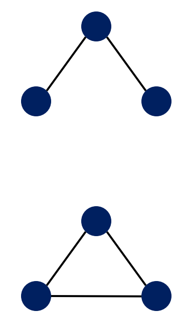
Example German biotech





Source: Ter Wal (2013)

Example German biotech

	ATTRIBUTE-RELATED EFFECTS	NETWORK STRUCTURAL EFFECTS
	Geographical proximity 	**Triadic closure** 

Source: Ter Wal (2013)

Example German biotech

	GEOGRAPHICAL PROXIMITY	TRIADIC CLOSURE
Phase 1 **EMERGENCE** ***Phase 2*** **GROWTH**	− ***Decrease over time***	+ ***Increase over time***

Source: Ter Wal (2013)

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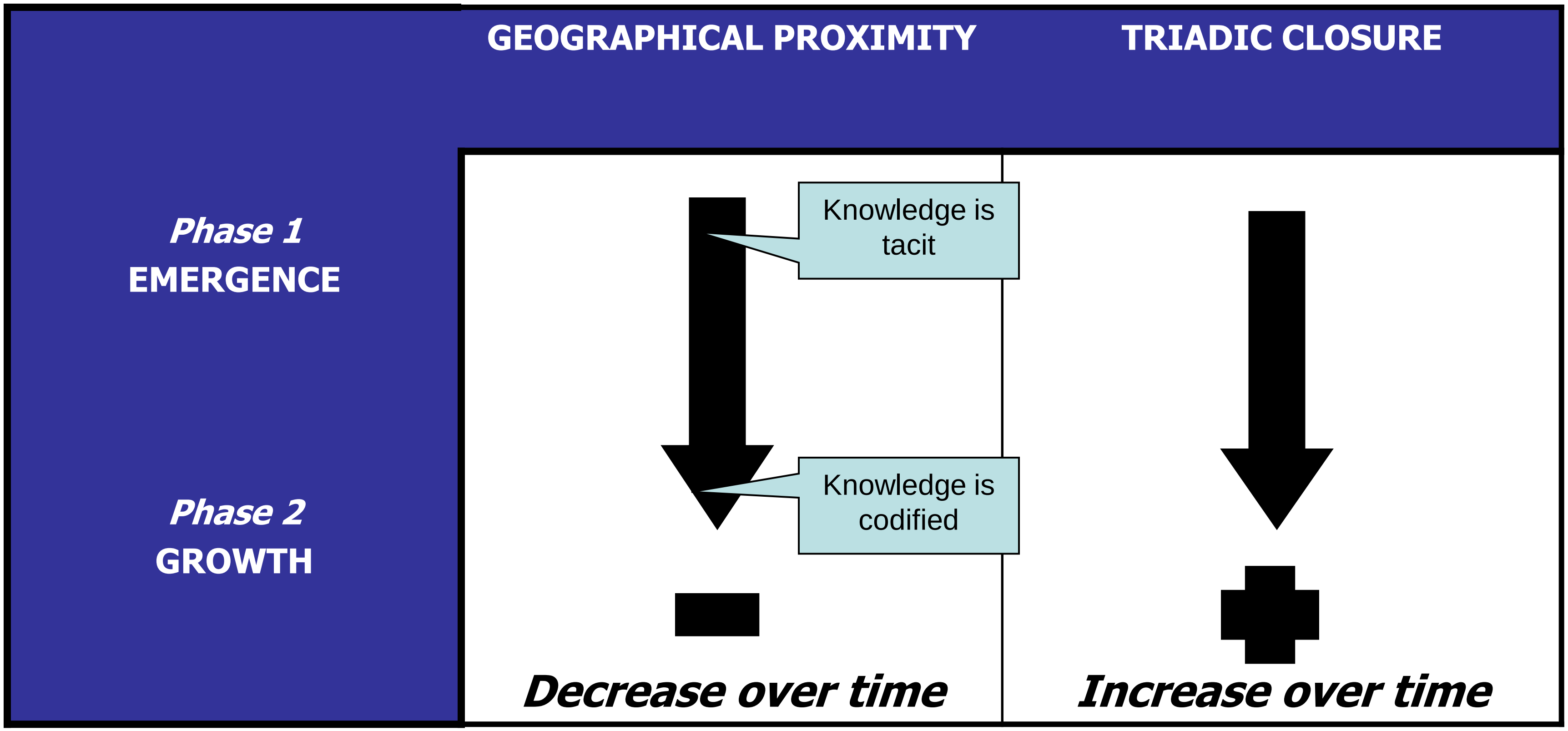
Example German biotech



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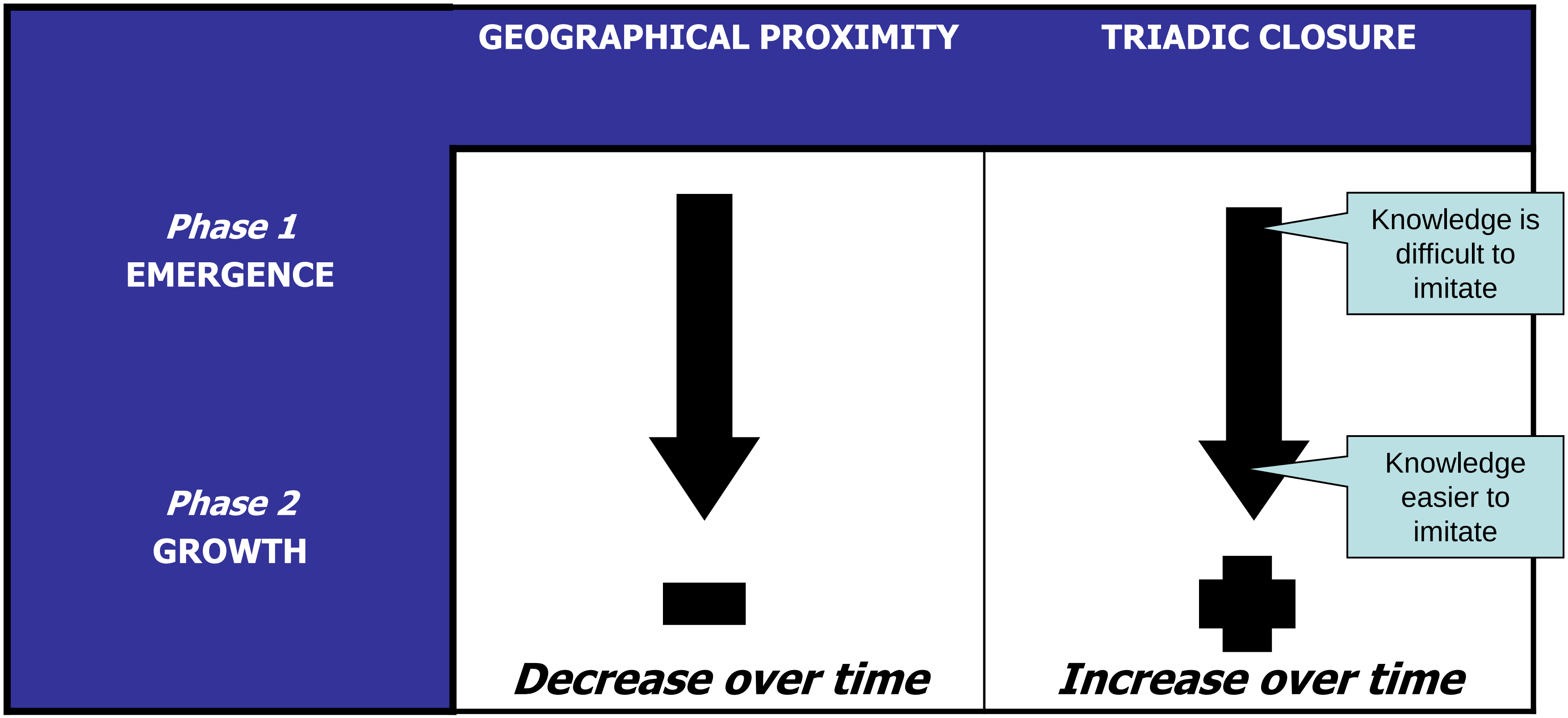
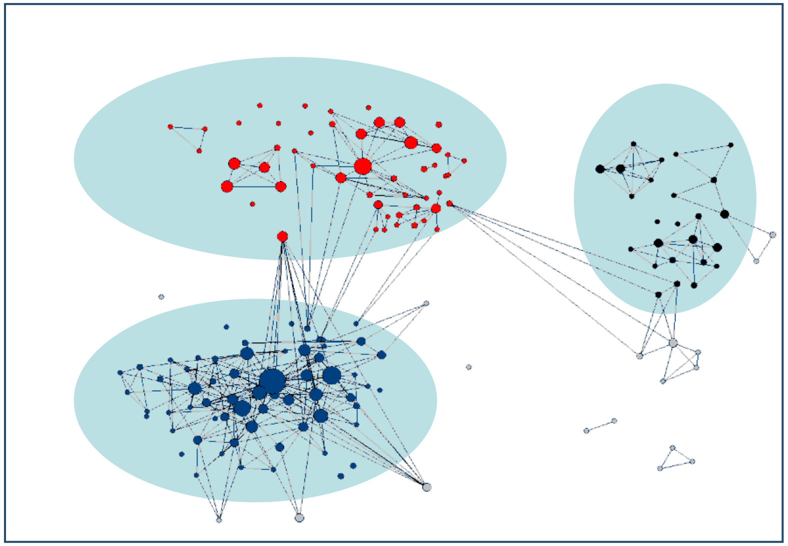
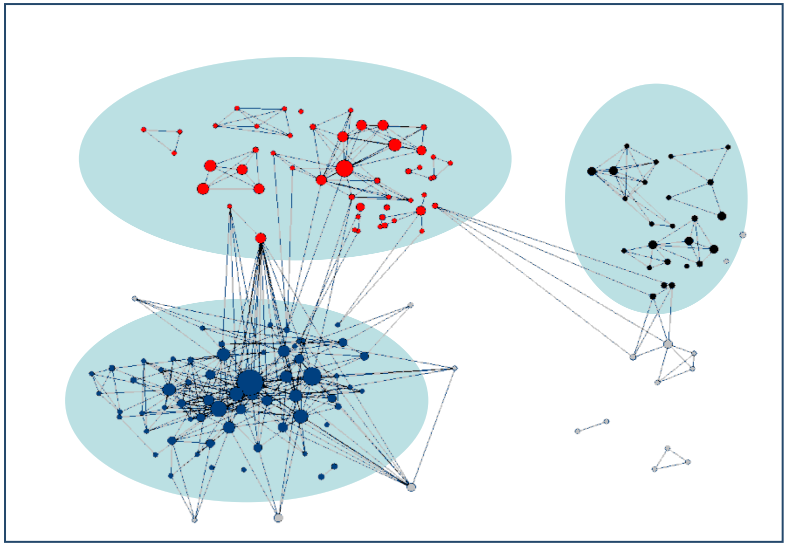
Example German biotech



Example German biotech



1980



1985

Triadic closure

Geographical proximity

Source: Ter Wal (2013)

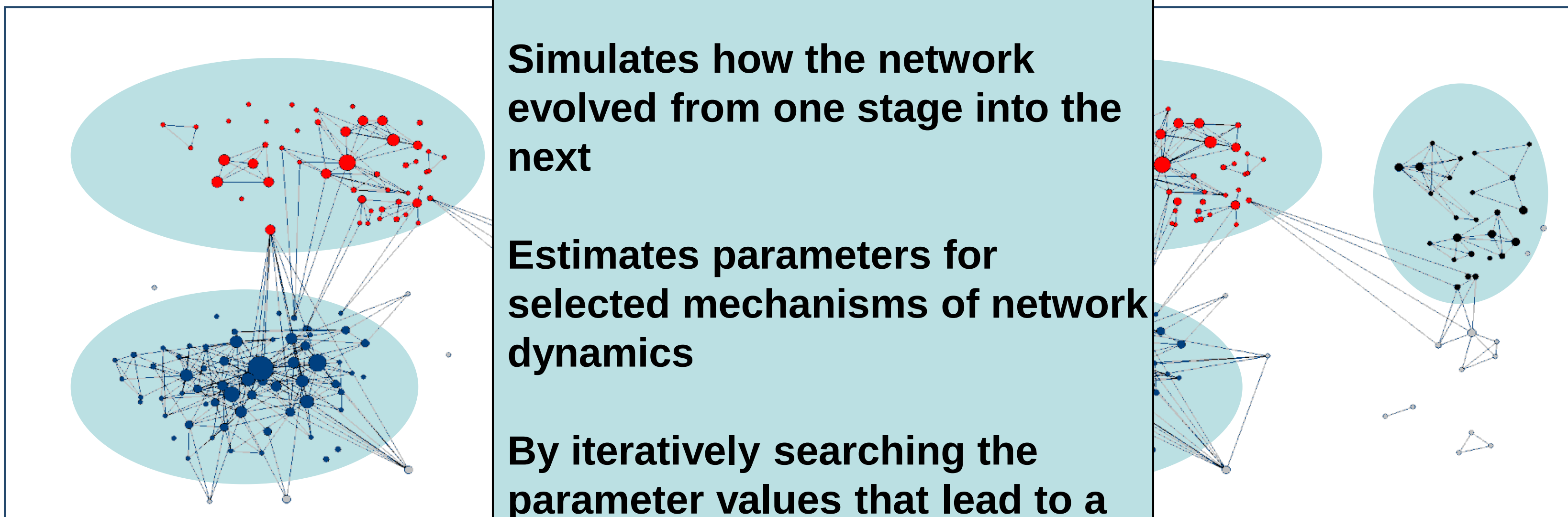
Example German biotech

SIENA:

Simulates how the network evolved from one stage into the next

Estimates parameters for selected mechanisms of network dynamics

By iteratively searching the parameter values that lead to a minimal deviation between generated and observed values of “target statistics”



1980

985

proxim

Source: Ter Wal (2013)

Example German biotech

	75-80	80-85	85-90	90-95
Number of nodes	49	96	122	214
Links created	17	64	45	87
Links dissolved	47	76	77	109
Links retained	50	78	111	178

	75-80	80-85	85-90	90-95
Rate of change	**1.4676** *** **(0,1866)**	**1.5069** *** **(0.1174)**	**1.0932** *** **(0.0937)**	**0.9571** *** **(0.0629)**
Degree	**-2.2926** *** **(0,0992)**	**-1.8339** *** **(0.0638)**	**-1.7881** *** **(0.0718)**	**-2.0348** *** **(0.0512)**
Geographical distance	**-0.0233** *** **(0,0031)**	**-0.0087** *** **(0.0021)**	-0.0042 (0.0034)	**-0.0094** *** **(0.0022)**
Triadic closure	**0.0785** *** **(0.0236)**	**0.4125** *** **(0.0425)**	**0.4812** *** **(0.0423)**	**0.6259** *** **(0.0546)**

Source: Ter Wal (2013)

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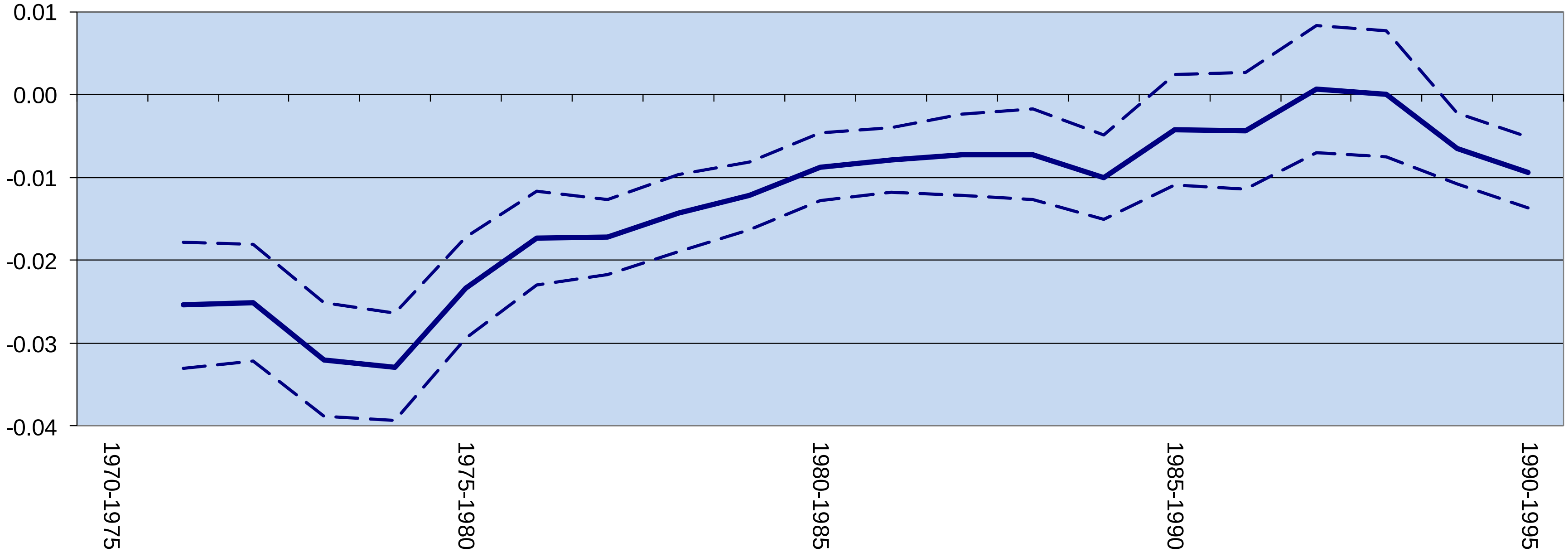
Example German biotech

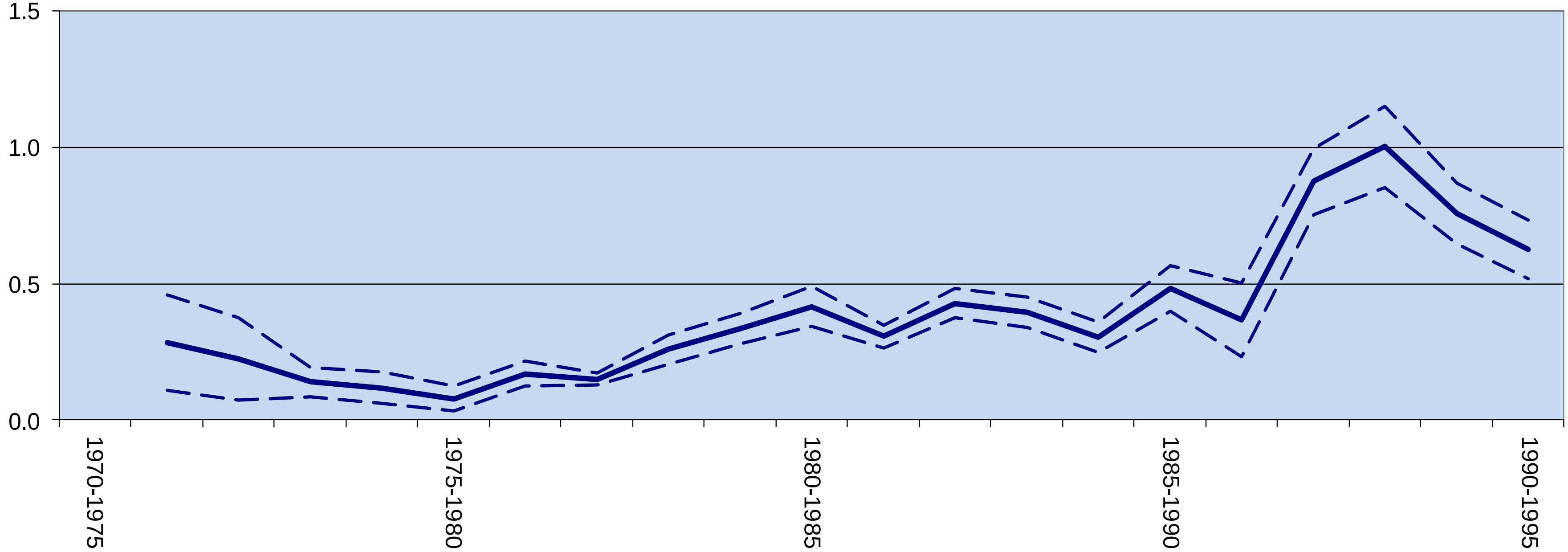
SIENA parameter estimate
“Geographical proximity” over time



Source: Ter Wal (2013)

Example German biotech

SIENA parameter estimate
“Triadic closure” over time



Source: Ter Wal (2013)

Example German biotech

- There is a **shift in emphasis** of how inventors select a collaboration partner, from exogenous, attribute-related mechanisms towards endogenous, network-structural effects.

- Whereas inventors initially tend to collaborate with **geographically proximate partners**, they increasingly direct their partner selection towards **closing pre-existing network structures**

- The dynamics of an industry network have a **spatial component** that is particularly relevant in the early stage of an industry, when knowledge is predominantly tacit.

- The **changing nature of knowledge** in an industry may have implications for how network dynamics unfold.

Source: Ter Wal (2013)

References

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